Makers Makin' It, Act II -- The Seequel

Project: p04 Global Grocery

Team: Tomatoes

Roster: Ashley Li (PM), Jason Chan, Sean Zheng, Maya Berchin

TARGET SHIP DATE: 2026-04-20

PROJECT DESCRIPTION:

A visitor to our site will learn about grocery prices throughout time and space. From this, they will be able to gauge trends and get the best deals, should they invent time travel and teleportation.

Upon logging in, users will be greeted with an interactive map and a search interface on the side. They will be able to input items to search for and find the optimal prices through time and space or explore the prices throughout the world by hovering over an interactive map. Looking for optimal prices will list the country containing a city with the lowest price in the whole database, and hovering over the map will show the user both the average price in the country and the lowest price in the country (both found from average prices across cities in that country). If searching for an optimal price using the search interface, the user can choose to optimize through time as well as space, but if interacting with the map, the user must choose a set time. The user can also save their favorite searches, which must include an item and may also include an year/month. Users can load their favorite searches to quickly search again.

PROGRAM COMPONENTS + EXPLANATION:

Python files

- *__init__.py*: the main file; serves app
- *data_setup.py*: handles parsing of CSV file and creation of Sqlite3 database tables
- *data.py*: handles the data in the Sqlite3 grocery database and the user data (all stored in *data.db*)

Templates

- *login.html*: the user will be directed onto the login page first. They will be redirected to the homepage once they are logged in (automatic if they are already logged in). If they don't have an account, they can be redirected to register.
- *register.html*: the user will be able to register. They will be redirected to the homepage once they do this.
- *home.html*: the homepage, which will have our interactive map and search interfaces.

CSS files

- *N/A*: we're gonna stick to our FEF

JavaScript files

- *map.js*: handles all the javascript we need for our interactive map

Other components

- *map.svg*: SVG used for our interactive map

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DATABASE ORGANIZATION:

Database Type:

Relational : We're taking data from a CSV file, so a table fits this dataset well. We also feel more familiar with SQLite3 than with a document one.

Tables:

users

Variable Type	Variable Name	Variable Attribute(s)
string	username	PK NOT NULL
string	password	NOT NULL

groceries

Variable Type	Variable Name	Variable Attribute(s)
string	city	NOT NULL
string	country	NOT NULL
string	name	NOT NULL
real	price	NOT NULL
integer	month	NOT NULL
integer	year	NOT NULL

favs

Variable Type	Variable Name	Variable Attribute(s)
string	search_id	PK NOT NULL
string	user	NOT NULL
string	item	NOT NULL
integer	month	
integer	year	

DATA VISUALIZATION LIBRARY:

We plan to use the D3 React Data Visualization Library, specifically the method of choropleth map, where different parts of the map are colored based on certain values (like price). For generating the map, we will be using an SVG for interactiveness.

DATA EXTRACTION:

To extract grocery data from the backend, we will first create a Flask route that queries the SQLite database based on the user's selected time and item, returning the price results formatted as a JSON response. This response would be passed to our JS file using the asynchronous fetch() API and will be parsed into a usable JavaScript object using the .json() method. The JS file will have a function that uses this parsed data to dynamically update the colors of our SVG map, while keeping specific details (like price) in memory for our sidebar whenever a specific country is clicked.

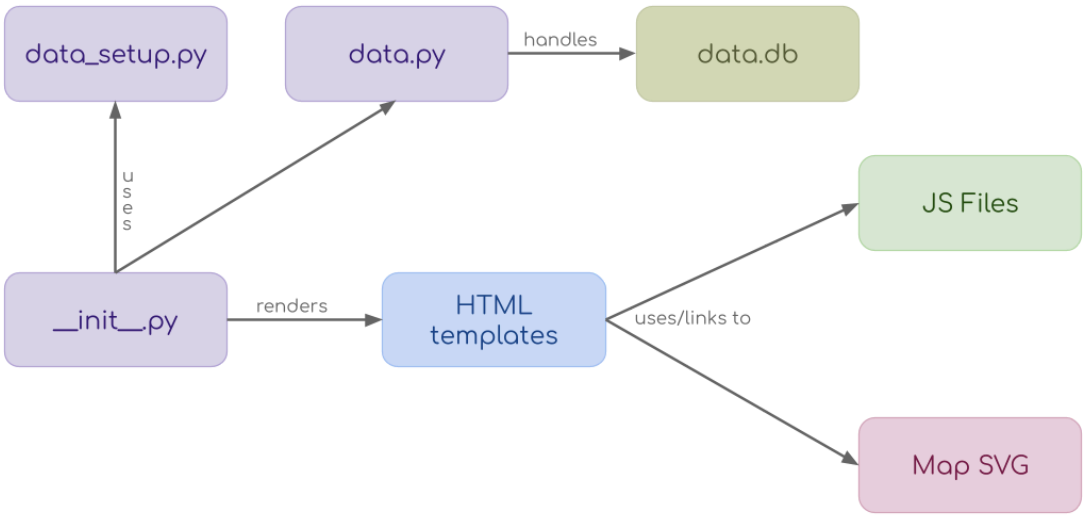
APIs:

We are not planning on using any external APIs for this project. Our data will be sourced from a CSV we downloaded.

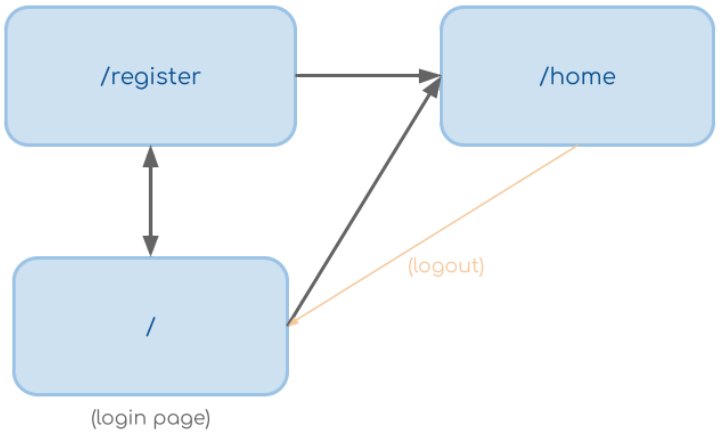
FRONT-END-FRAMEWORK:

Tailwind : 3/4 Devos are most familiar with Tailwind, and the other also feels comfortable with it.

COMPONENT MAP:



FRONT END DIAGRAM:



TASK DELEGATION:

Task	Devo(s)	Deadline	Status
Grocery Database handling	Maya	04-01w	done
User Database handling	Maya	03-31t	done
Map Generation (SVG)	Sean	04-21t	done
Map Interactive Features	Sean	04-21t	done
Helper Functions for search	Jason	04-20m	done
Homepage python	Maya	04-17f	done
Register page + backend	Jason	04-17f	done
Login page + backend	Jason	04-17f	done
Logout functionality	Jason	04-16r	done
Register page frontend	Ashley	04-20m	done
Login page frontend	Ashley	04-20m	done
Homepage frontend	Ashley	04-22w	done
Devlog	All	04-22w	done